

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

KINETIC THEORY

LECTURE-01

PHYSICS

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Topics To Be Covered

- 1** Equation of state of a perfect gas
- 2** Work done in compressing a gas
- 3** Kinetic theory of gases-assumptions
- 4** Concept of pressure
- 5** Expression for pressure exerted by a gas

Kinetic Theory of Gases

Kinetic Theory

Equation of state of a perfect gas. Work done in compressing a gas.

Kinetic theory of gases - assumptions, concept of pressure. Expression for pressure exerted by a gas. Kinetic interpretation of temperature; rms speed of gas molecules; degrees of freedom, law of equi-partition of energy (statement only) and application to specific heat capacities of gases; concept of mean free path, Avogadro's number.

KTG (Kinetic theory of gas molecule)

①



Points | Postulates

- ① Gas is the state of matter
- ② Gas is made up of Gas Molecule
- ③ Gas molecules are always in the State of **Random**

p4a **Motion**

④ Gas molecule have velocity (v), and hence
They have $K.E = \frac{1}{2}mv^2$



KTG (Kinetic theory of gas molecule)

2

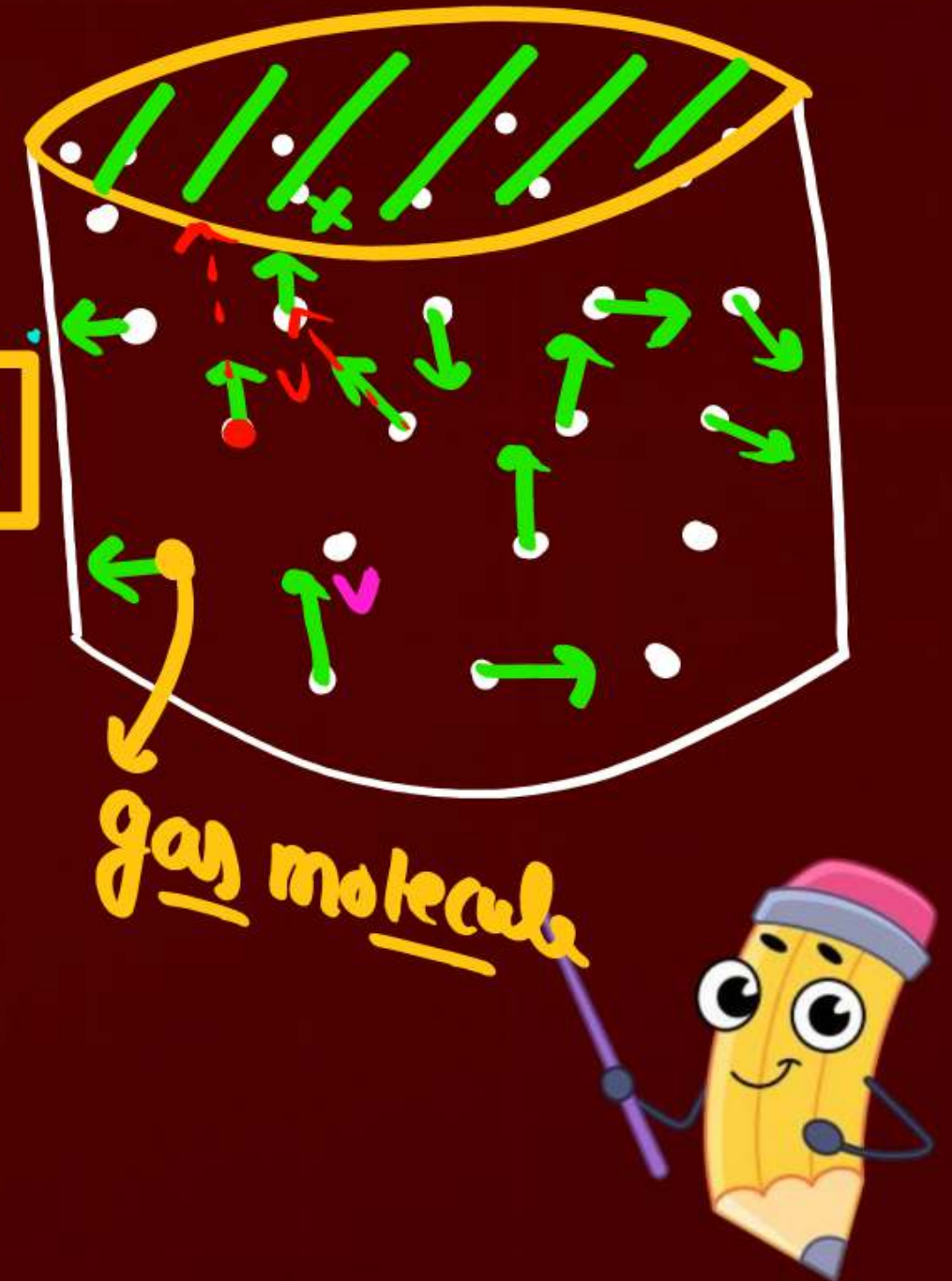


Points / Postulates

⑤ gas molecule, collide with each other and with the wall of Container

⑥ Their collision is **Elastic** **Collision**

They do not lose the energy during collision



KTG (Kinetic theory of gas molecule)

3



Points / Postulates

⑦ Gas molecule Exert the **Pressure** on the wall of the container

Pressure $\rightarrow P \propto \rho$ — ① $\rightarrow$ Density
 $P \propto v^2$ — ② $\rightarrow$ velocity

$K \rightarrow$ pressure constant

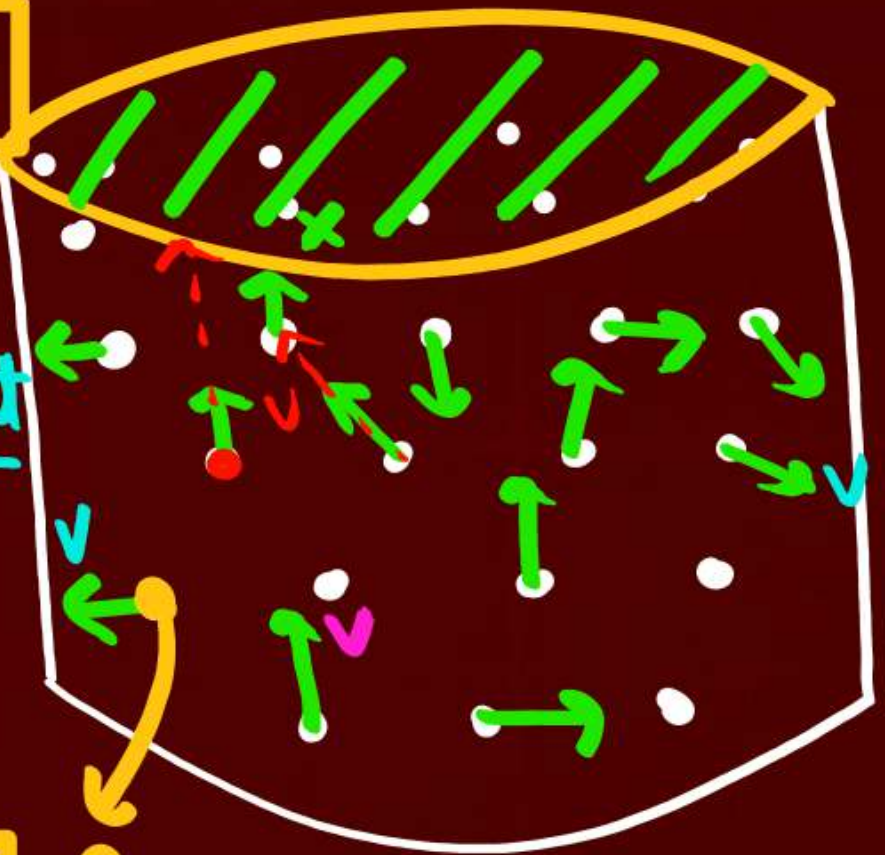
$$K = \frac{1}{3}$$

$$P \propto \rho v^2$$

$$P = K \rho v^2$$

$$P = \frac{1}{3} \rho v^2$$

expression for pressure



gas molecule



KTG (Kinetic theory of gas molecule)

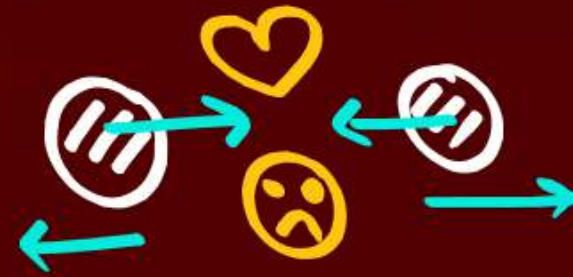
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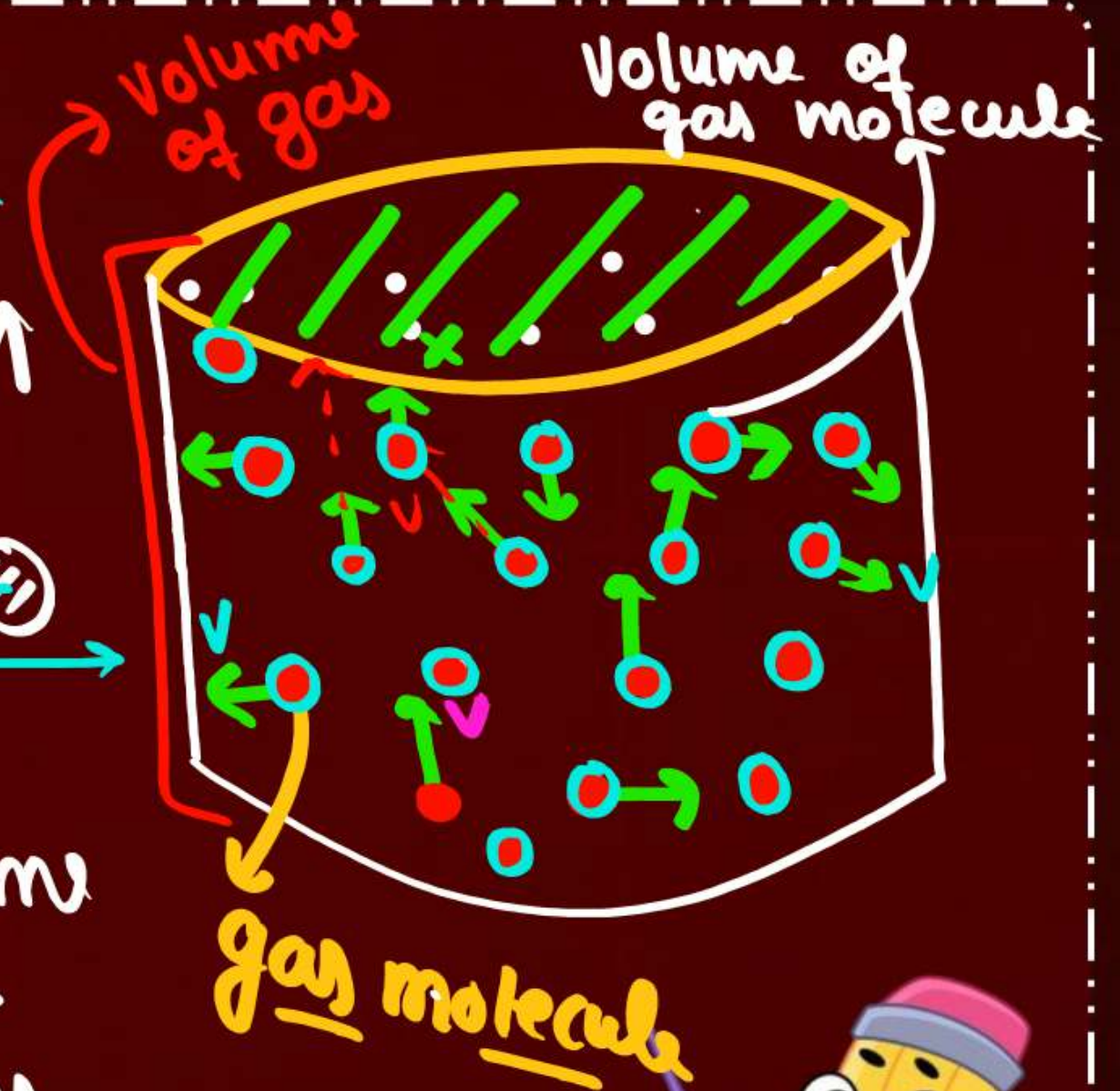
Points / Postulates

⑧ on increasing the temperature $\uparrow\uparrow$
Temp $\uparrow\uparrow$ $V \uparrow\uparrow$ K.E $\uparrow\uparrow\uparrow$ $P \uparrow\uparrow\uparrow$

⑨ There is no interaction b/w the gas molecules



⑩ Volume of gas $\gg \gg \gg \gg$ Volume of gas molecule

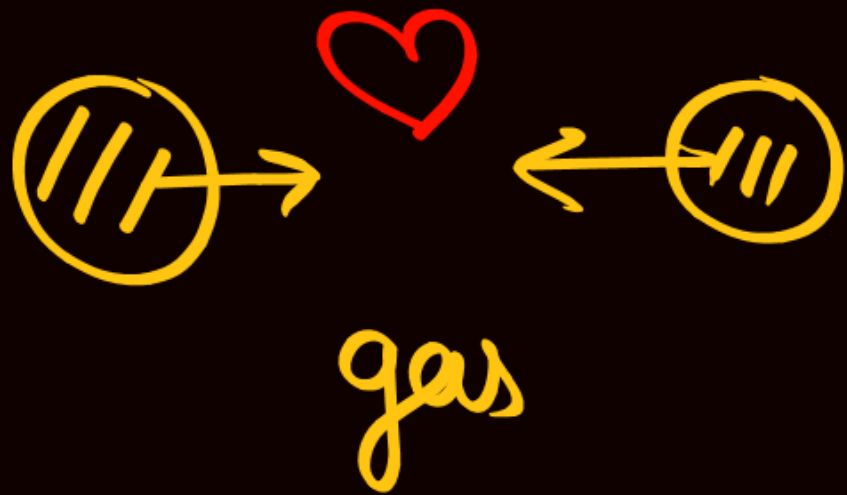


Q Which **two** **postulate** of **KTG** is wrong?

2 Mark 2023
2021
2017

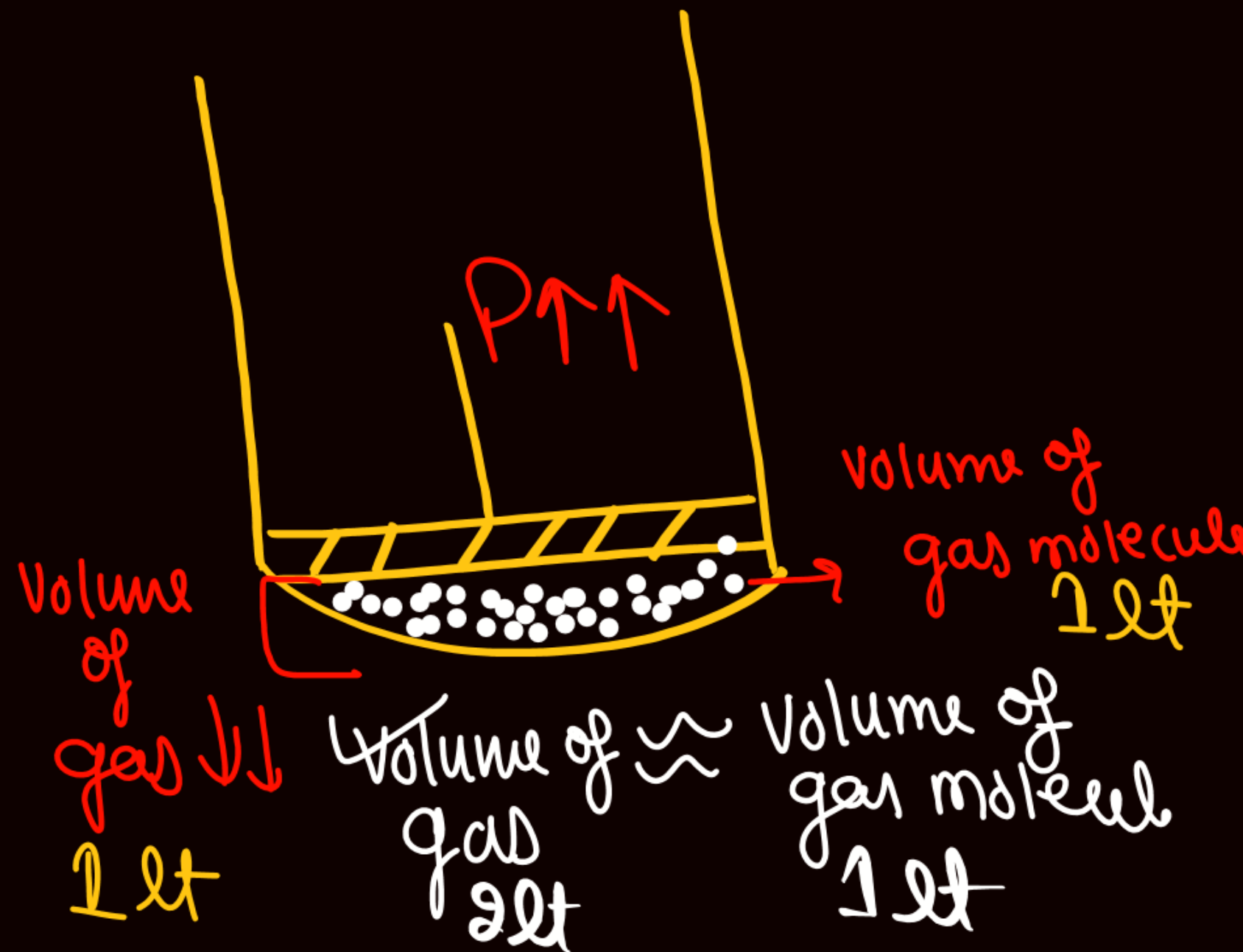
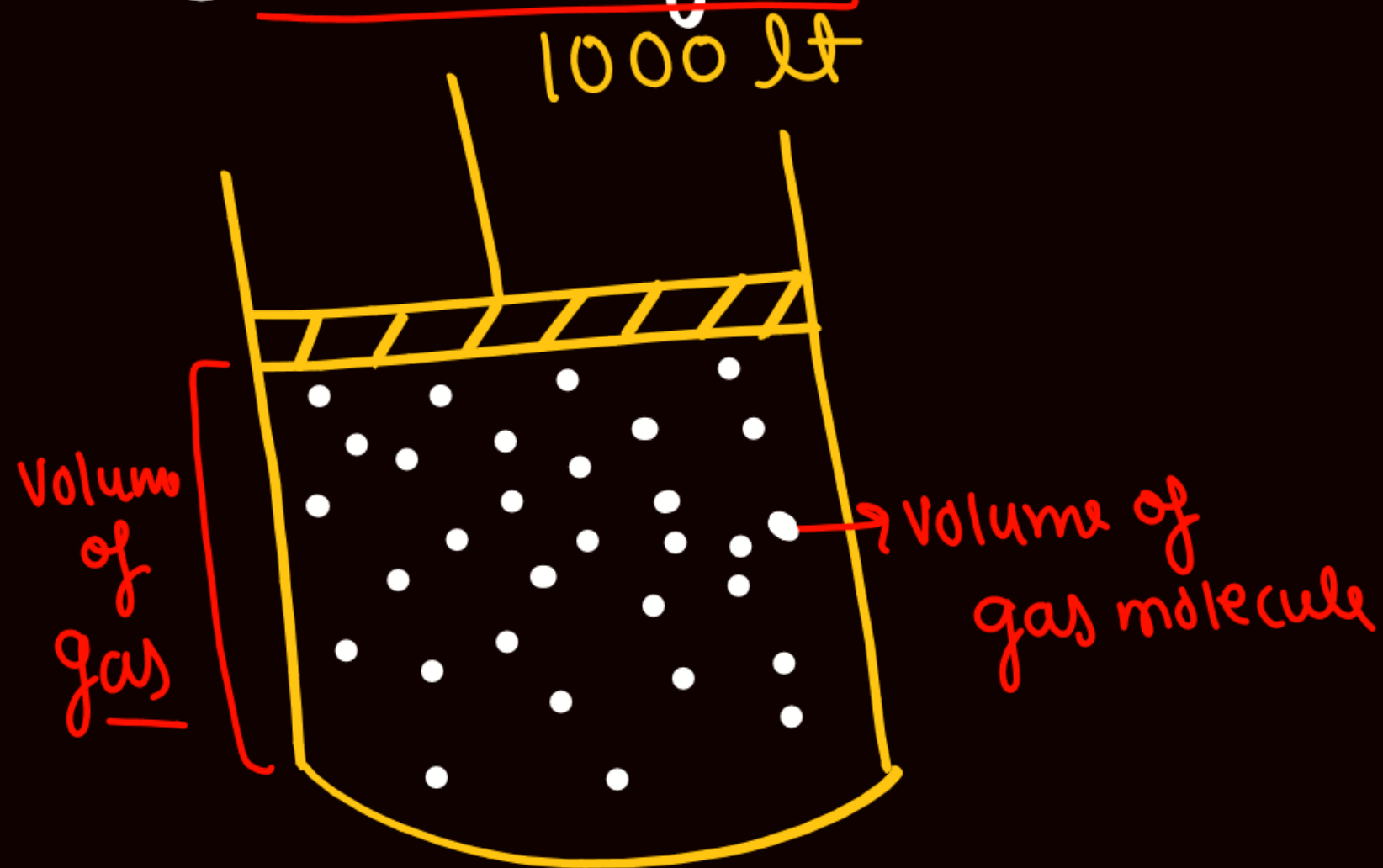
⑤

Ans ① There is **no interaction** b/w the gas molecule



② Volume gas > > > > Volume of gas molecule

⑥



Relationship blw pressure and K.E

2024 3 mark

⇒ As velocity ↑↑↑ K.E ↑↑ Pressure ↑↑

Maths

KTG

$$P = \frac{1}{3} \rho v^2$$

$$\rho = \frac{m}{V}$$

mass

volume

$$P = \frac{1}{3} \frac{m}{V} v^2$$

velocity

$$PV = \frac{2}{3} \left(\frac{1}{2} mv^2 \right)$$

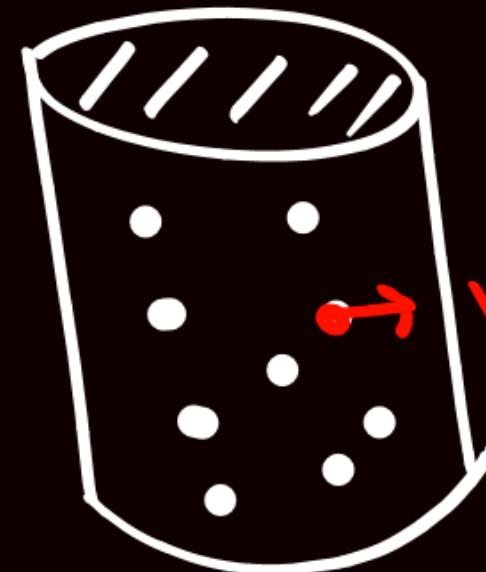
K.E

$$PV = \frac{1}{3} mv^2$$

$$PV = \frac{1}{3} \frac{2}{2} mv^2$$

$$\uparrow PV = \frac{2}{3} K.E \uparrow$$

$$P = \frac{2}{3} \frac{K.E}{V \rightarrow \text{Volume}}$$



V ↑ K.E ↑ P ↑

Temp ↑

$$\uparrow P = \frac{2}{3} \frac{K.E}{V \rightarrow \text{Volume}} \uparrow$$

⑦

Relationship b/w K.E and Temperature

3 Mark

2025
2024 (8)
2023
2021

As we increase the temperature $\uparrow\uparrow\uparrow$

$V \uparrow\uparrow$ K.E $\uparrow\uparrow$

$$PV = \frac{1}{3} \cdot 2 \cdot mv^2$$

Math
KTCG
 $P = \frac{1}{3} \rho v^2$

$\rho = \frac{m}{V}$
velocity v
Volume V

$$P = \frac{1}{3} \frac{m}{V} v^2$$

$$PV = \frac{1}{3} mv^2$$

$$PV = \frac{2}{3} \left(\frac{1}{2} mv^2 \right)$$

$$nRT = PV = \frac{2}{3} K.E$$

$$nRT = \frac{2}{3} K.E$$

$$\frac{3}{2} nRT = K.E$$



$V \uparrow$ K.E $\uparrow$

Temp $\uparrow$

$$\uparrow K.E = \frac{3}{2} nRT \uparrow$$

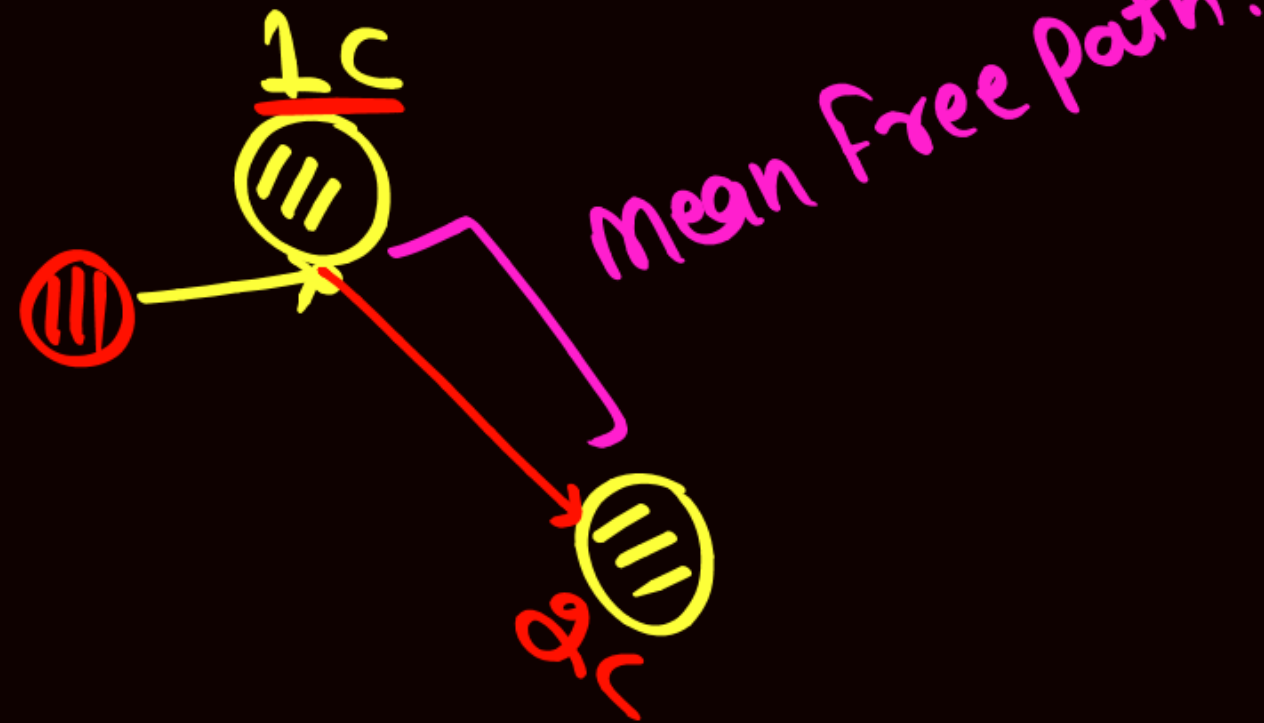
$$\uparrow K.E = \frac{3}{2} nRT \uparrow$$

Mean Free Path

2019

2024

:- the distance between two successive collision



SUMMARY

$KTG \quad P$

2

$P \rightarrow K \cdot \epsilon$

$K \cdot \epsilon \rightarrow T$

MFP

HOMEWORK



Copy → (cp)

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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